HIREFLOW PRODUCTION ATS PLATFORM v4.0.0

TAILOR MY RESUME STUDIO

COMPLETE CONTEXTUAL BRIEF, ARCHITECTURE, CONNECTIONS & DATA DICTIONARY

EXECUTIVE SYSTEM INVARIANTS
100% Client-Side Reactive SaaS Engine | Zero Runtime Network Egress | Strict Factual Non-Fabrication Truth Gate | Google XYZ Standards Elevation | Native Vector PDF & DOCX Binary Generators

1. EXECUTIVE OVERVIEW & DESIGN PHILOSOPHY

Tailor My Resume (/tools/tailor) is HireFlow's flagship accomplishment elevation studio. It provides engineering and technology candidates with an interactive, evidence-grounded environment to align their resume with specific Job Descriptions (JDs) while maintaining strict factual truth.

- Deterministic Factual Truth: Unlike standard generative AI which hallucinates metrics, dates, and unverified competencies, HireFlow strictly prohibits inventing numbers. Every bullet is cross-verified against an immutable Candidate Fact Ledger.
- Google XYZ Standard: Systematically restructures weak task statements into executive-grade accomplishment bullets: Accomplished [X] measured by [Y] by doing [Z].
- Zero Network Egress: Ingestion, NLP parsing, knowledge graph matching, layout height simulation, DOCX compilation, and vector PDF rendering occur entirely inside the user's browser.
- 1-Page Density Sentinel: Real-time calculation of vertical pixel height (1056px Letter / A4) prevents single-line spills onto page 2 and auto-tunes typography.

2. ECOSYSTEM INTEGRATION & DIRECTED NAVIGATION

Tailor My Resume functions as the central execution studio across HireFlow, receiving qualified data from discovery tools and feeding into verification pipelines:

Inbound Deep Links & Parameter Handlers

- ATS Resume Roaster (FixTheseFirst.tsx): When high-ROI gaps are found, 1-click handoffs route to /tools/tailor?focus=<item_title> (missing mandatory requirements) or /tools/tailor?mode=upgrade&bullet=<bullet_title> (weak bullet elevation).
- ATS Skills Tab (SkillsTab.tsx): Direct link /tools/tailor?skill=<canonical_skill> launches the studio pre-focused on bridging that exact skill into experience or project entries.
- Tools Hub (ToolsHubPage.tsx): Positioned prominently in Group A "Application & Outreach" with badge "Truth-Checked Suggestions" and tags [Truth Check, Diff Editor, Version History].
- Compare Job Descriptions (HeadToHeadView.tsx): Dual action buttons "Tailor for Role A" and "Tailor for Role B" allow candidates to fork role-specific branches from one source resume.
- Placement Pathways (plan.ts & StudentPlacementPathway.tsx): Direct call-to-action from readiness diagnostic to the "Single-Page Technical Fresher Resume Tailoring" milestone.
- Global Navigation (LocalhostNavigator.tsx & ProfileResumeChip.tsx): Shared profile context synchronizes candidate text across ATS Roaster, TalentLens, and Tailor.

Outbound Directed Handoffs

- Build Proof Project Briefs (/tools/build-briefs?query=<skill>): When a mandatory requirement is completely missing and cannot be honestly claimed on the resume, the studio links the candidate to a hands-on project brief to build verifiable proof before applying.

3. APPLICATION ARRANGEMENT & WORKSPACE LAYOUT

The studio interface is divided into a high-productivity split-screen layout with synchronized live rendering:

1. Top Navigation Bar (DarkTailorHeader.tsx)

- Dynamic Real-Time Score Badge: Computes baseline vs upgraded score (0-100%) reflecting power verb coverage (40%), metric density (40%), and passive pattern elimination (20%).
- Quick Archetype Presets: Instant toggle between Fresher SDE (Jeswanth Kumar) and Cybersecurity Specialist (Alex Morgan CISSP).
- Multi-Format Export Suite: Direct downloads for .DOCX (Word binary), .PDF (vector binary), .TXT (ATS plain text), .MD (Markdown), and .JSON (reactive backup).
- Make It Better Trigger: Initiates global Google XYZ transformation modal.

2. Left Column: Handmade Resume Editor (HandmadeResumeEditor.tsx)

- Filter Tabs: Instant subsection filtering (All, Profile, Summary, Experience, Projects, Skills, Education, Requirements).
- Live Search Bar: Highlights matching terms and filters entries dynamically.
- Auto-Fit 1 Page Sentinel: One-click algorithm that adjusts margins, font tokens, and line spacing to force content onto 1 page.
- 7 Dedicated Section Editors: WorkExperienceSection, ProjectsSection, SkillsSection, EducationSection, SummarySection, JobRequirementsSection, and CandidateProfileSection.

3. Right Column: Live Dark Paper Canvas (DarkPaperPreview.tsx)

- Real-time A4 / Letter simulation with print typography, border styling, zoom scaling (80%-110%), and overflow alerts.

4. CORE DATA MODELS, SCHEMAS & FACT LEDGER

Every operation operates on strictly-typed data models defined in src/lib/tailor/types.ts:

Interface / Model	Key Fields & Types	Architectural Purpose
CandidateFact	factId, sourceSection, rawText, normalizedConcepts, extractedMetrics, hash	Immutable ledger of candidate history preventing AI hallucinati
JobRequirement	id, text, category, priority (MANDATORY|PREFERRED), weight, normalizedConcepts	Extracted target JD with numerical scoring weig
RequirementAssessment	requirementId, status (DIRECT|TRANSFERABLE|PARTIAL|MISSING), supportingFactIds	Mapping between candidate facts and targe
BulletTransformation	bulletId, factId, originalText, tailoredText, actionVerb, preservedMetrics	Before-and-after bullet transformation record with audit ratio
LayoutSettings	template, paperSize, fontSize, margins, lineSpacing, accentColor	Visual layout configuration controlling 1-page fit and export s
TruthGateResult	passed, score, unsupportedClaimsCount, preservedMetricsCount, blockingIssues	Audit blocking export if invented facts are det

5. INTELLIGENCE & CLASSIFICATION ENGINES

Candidate Archetype Classifier & Dynamic Section Ordering (strategyPlanner.ts)

The strategy planner dynamically reconfigures the section hierarchy based on candidate seniority and target domain:

Archetype	Recommended Section Ordering	Target Persona / Strategy
student_fresher	Education -> Projects -> Skills -> Experience -> Certs	Entry-level campus hires; prioritizes academic excellence
cybersecurity_specialist	Security Summary -> Certifications -> Experience -> Skills -> Labs	Infosec roles; prioritizes CISSP/CEH trust signals and SO
academic_researcher	Education -> Publications -> Experience -> Methodologies	R&D / Ph.D. roles; prioritizes peer-reviewed publications a
experienced_executive	Executive Summary -> Experience -> Core Competencies -> Education	Leadership / VP roles; frames P&L responsibility, team sc
technical_specialist	Technical Skills -> Work Experience -> Projects -> Education	Mid/Senior SDEs; front-loads technical stack for immedia

Make It Better Engine & Weak Opener Replacements (makeItBetterEngine.ts)

- Weak Opener Elimination: Automates replacement of 12 passive patterns (e.g. "Responsible for" -> "Spearheaded", "Worked on" -> "Engineered", "Assisted with" -> "Co-engineered", "Handled" -> "Streamlined").
- Context-Sensitive Power Verbs: Selects high-impact verbs based on technical domain: Performance/Latency -> "Optimized", Leadership -> "Spearheaded", Security -> "Fortified", Cloud/DevOps -> "Automated".
- Strict Metric Locking: 100% of candidate numeric metrics are extracted and preserved. No synthetic numbers are ever injected.

Requirement-Aware Accomplishment Synthesis (accomplishmentGenerator.ts)

- 7 Domain Suggestion Packs: Dynamically synthesizes Google XYZ accomplishment bullets across Scale & Latency, Architecture & APIs, Frontend & UX, Data & DB, Cloud & DevOps, Security & Auth, and Quality & Testing.

6. TRUTH GATE & NON-FABRICATION INVARIANTS

HireFlow operates a dual-sentinel truth gate system to enforce zero fabrication at runtime:

SENTINEL 1: INTERACTIVE TRUTH GATE (SRC/LIB/TAILOR/ENGINE/TRUTHGATE.TS)
* Invented Metric Detection: Scans every number, latency value, and dollar figure. If a number did not exist in the candidate fact ledger, it triggers a CRITICAL blocking issue. * Unverified Skill Scanner: Flags any technology not present in the candidate profile with a WARNING until candidate confirmation is received.

SENTINEL 2: HARSH INVARIANT GATE (SRC/ENGINE/TAILOR/TRUTHGATE.TS)
* Claim-Strength Ladder (CLAIM_STRENGTH_LADDER): Prohibits escalating action verb strength beyond source ownership (e.g., cannot escalate "assisted" (Tier 1) to "architected" (Tier 5)). * Keyword Mirroring Cap (TAILOR_MIRRORING_LIMITS): Limits any target keyword to max 2 mentions across tailored bullets to prevent ATS spam penalties. * Deterministic Quality Fallback: If any fatal violation occurs, the system automatically falls back to original bullet phrasing.

7. CANDIDATE ACTIONS & COMPLETE CHANGE CAPABILITIES

The candidate retains full interactive agency over every aspect of their resume:

- 1. Reorder & Toggle Sections: Move sections up/down or hide non-essential sections (e.g., hide Certifications for freshers) without deleting data.
- 2. In-Place Bullet Elevation: One-click "XYZ Elevate" on any bullet point to convert passive duties into metric-driven accomplishments.
- 3. Metric Placeholders: Injects structured prompts ([X%], [N ms], [M requests/sec]) guiding candidates to provide verifiable data.
- 4. Skills Taxonomy Management: Add or remove skill chips across 7 standardized categories.
- 5. 1-Page Auto-Fit Sentinel: Single-click optimization of layout margins (0.4in-0.65in), font tokens, and spacing.
- 6. Multi-Template Switcher: Switch seamlessly between Classic, Modern, Compact, and Executive visual styles.
- 7. Version Snapshots & Privacy Purge: Take versioned snapshots with 1-click rollback, or trigger a complete privacy wipe that permanently clears LocalStorage.

8. MULTI-FORMAT EXPORT SUITE & SENTINELS

All exports are generated locally via pure client-side libraries with pre-export validation:

Export Formats

- Vector PDF (.pdf): Authentic binary PDF compiled via pdf-lib with embedded standard typography (Helvetica/Times-Roman), page break calculation, and glyph sanitization.
- Word Binary (.docx): True Microsoft Word binary built via docx library featuring ATS single-column formatting, standard headings, and bullet margins.
- Plain Text (.txt) & Markdown (.md): Clean ASCII formatted text formatted specifically for pasting into ATS web forms.
- JSON State Backup (.json): Machine-readable dump of the full active project state.

Quality & Sentinel Audits

- Layout & Height Sentinel (layoutSentinel.ts): Evaluates vertical height against 1056px capacity, flagging single-line spill risks.
- Pre-Export Formatting Checker (formattingChecker.ts): Scans for broken glyphs (Unicode replacement chars), widow/orphan lines (<3 words on final line), and missing date ranges.

9. CODEBASE DIRECTORY & SYMBOL REFERENCE

File Path	Key Symbols	Module Function
src/lib/tailor/store/tailorStore.ts	useTailorStore, createDefaultProject	Reactive Zustand state, pipeline stages, auto-fit.
src/lib/tailor/engine/makeItBetterEngine.ts	elevateBulletPoint, elevateResumeToBestStandard	Google XYZ bullet transformer and scoring policy.
src/lib/tailor/engine/truthGate.ts	verifyFactualIntegrity	Metric preservation and unverified skill blocking.
src/engine/tailor/truthGate.ts	evaluateTruthGate	Claim-strength ladder and keyword mirroring cap.
src/lib/tailor/engine/strategyPlanner.ts	planTailoringStrategy	Archetype classification and section reordering.
src/lib/tailor/matching/evidenceMatcher.ts	matchCandidateToJob	4-tier requirement assessment and transferability.
src/lib/tailor/engine/accomplishmentGenerator.ts	generateUniqueAccomplishments	Requirement-mapped accomplishment suggestions.
src/lib/tailor/export/pdfExporter.ts	generatePdfResume, downloadResumeAsPdf	Client-side vector PDF generation via pdf-lib.
src/lib/tailor/export/docxExporter.ts	generateDocxBlob, downloadResumeAsDocx	ATS-compliant Word document binary generator.
src/lib/tailor/export/layoutSentinel.ts	evaluateLayoutHeight	1056px page height calculator and spill risk detector.
src/lib/tailor/export/formattingChecker.ts	runPreExportFormattingChecks	Glyph audit, widow/orphan detector, date validator.
src/features/suite/tailor/TailorResumePage.tsx	TailorResumePage	Top-level page with split editor/canvas layout.
src/features/suite/tailor/components/HandcraftedResumeEditor.tsx	HandcraftedResumeEditor	Subsection 2-column editor with filter tabs.
src/features/suite/tailor/components/DarkPaperPreview.tsx	DarkPaperPreview	Live A4/Letter paper canvas with print CSS sync.